DroneSSL: Self-Supervised Multimodal Anomaly Detection in Internet of Drone Things

Junaid Akram, Ali Anaissi, Wajdy Othman, Abdulatif Alabdulatif, *Member, IEEE*, and Awais Akram

***Abstract*—In this study, we introduce a pioneering framework, DroneSSL, that integrates the concept of spatial crowdsourcing with TinyML to enhance anomaly detection in the Internet of Drone Things (IoDT). This innovative approach leverages drones and unmanned ground vehicles (UGVs) for expansive data collection in environments that are typically inaccessible or hazardous, such as during Australian bushfire incidents. By employing lightweight machine learning models alongside advanced communication technologies, DroneSSL transcends traditional spatial-temporal data analysis methods. It efficiently processes multimodal data from diverse Points-of-Interest (PoIs), significantly improving the quality and speed of data collection and analysis. The framework's integration of a temporal feature extraction module with a Graph Neural Network (GNN) and its adaptable, scalable GNN architecture tailor DroneSSL for real-time operations in resource-constrained IoDT environments. Achieving an 89.6% F1 score, DroneSSL marks a substantial 4.9% improvement over existing approaches, highlighting its effectiveness in critical applications such as environmental surveillance and emergency response. This advancement not only showcases the potential of combining TinyML with spatial crowdsourcing for IoDT but also sets a new standard for efficient, scalable anomaly detection, paving the way for future innovations in IoT edge devices and environmental monitoring systems.**

***Index Terms*—Self-supervised learning, autoencoder, GNN, anomaly detection, drones, data fusion, TinyML.**

I. Introduction

In THE evolving landscape of consumer electronics, deeply intertwined with our daily lives, TinyML emerges as a pivotal technology reshaping how we interact with smart devices [1]. From smartphones to wearable technologies, the integration of TinyML is propelling these gadgets beyond their traditional roles, aligning with the Internet of Things (IoT) paradigm [2]. This progression has ushered in a new era where consumer electronics transcend individual functionality, becoming integrated elements of a broader, interconnected ecosystem [3].

The advent of the Internet of Drone Things (IoDT) epitomizes this technological leap, extending the reach of consumer electronics into the skies [4]. IoDT networks, encompassing sensor-laden drones, cater to a spectrum of applications, from environmental surveillance, such as Australian bushfire monitoring, to defense and healthcare [5], [6], [7], [8], [9], [10]. However, deploying IoDT systems, especially in remote or challenging environments, invites a range of disruptions and security risks, including signal interference and data breaches [11], [12]. These systems are further strained by inherent limitations in resources like energy and bandwidth [13].

Integrating the concept of spatial crowdsourcing with IoDT represents a groundbreaking approach to overcoming these challenges, particularly in environmental monitoring contexts such as Australian bushfire management [4], [5]. Spatial crowdsourcing, when combined with IoDT, leverages drones and unmanned ground vehicles (UGVs) for expansive, real-time data collection in areas that are typically hazardous or unreachable by humans. This synergy not only addresses the limitations of traditional human-operated spatial crowdsourcing by covering larger areas with greater efficiency but also enhances the quality and speed of data collection from Points-of-Interest (PoIs) through advanced communication technologies like WiFi/5G [8], [10]. The incorporation of air-ground non-orthogonal multiple access (AG-NOMA) techniques further optimizes data transmission between drones and UGVs, facilitating high-quality data relay and decoding [9].

One of the critical aspects of IoDT, enhanced by the integration of spatial crowdsourcing, is the development of lightweight models for anomaly detection. These models are essential for identifying deviations in data flow patterns, which could indicate bushfire developments or other environmental anomalies. Current research often overlooks the spatial location information among drones and the correlation of data across various modes, a gap that our approach, DroneSSL, aims to fill. DroneSSL, a unique lightweight self-supervised learning framework designed for IoDT, incorporates TinyML principles with spatial-temporal data analysis. This not only advances anomaly detection but also significantly improves the functionalities of consumer electronics within the IoDT domain. By leveraging spatial crowdsourcing, DroneSSL can process and analyze data from a broader area more efficiently, enhancing the system's adaptability and performance in critical applications like bushfire monitoring.

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TABLE I
COMPARISON OF ANOMALY DETECTION METHODS IN IoDT

Method/ Reference	**Technology Used**	**Strengths**	**Limitations**	**Applicability to IoDT (e.g., Bushfire Monitoring)**
Fei et al. [14]	Euclidean Clustering	Simple, effective clustering	Lacks spatial correlation analysis	Limited applicability due to lack of spatial analysis
Ding et al. [15]	Correlation Analysis	Effective for high-dimensional time series	Does not consider multimodal scenarios	Suitable for temporal analysis, but limited in multimodal contexts
Li et al. [16]	GAN with LSTM	Captures temporal correlations	Complex implementation	Effective for diverse data flow analysis, such as in bushfire patterns
Park et al. [17]	VAE and LSTM	Efficient in multimodal data processing	Requires large datasets for training	Useful for analyzing multimodal data from drones
Yu et al. [18]	CNN and Incremental Learning	Adaptable to new data or classes	Requires retraining for new scenarios	Good for evolving IoDT scenarios like changing bushfire conditions
Zhang et al. [19]	Self-Supervised Learning with Memory Fusion	Learns both common and special features	May have high computational overhead	Suitable for large-scale data processing in IoDT
Zhao et al. [20]	Semantic Representation Learning	Effective in spatial feature capture	Might be complex to implement	Highly applicable for spatial analysis in IoDT
Xu et al. [21]	Minimax Strategy with Attention Mechanism	Enhances data difference detection	May not scale well with very large datasets	Useful for detailed anomaly detection in complex IoDT environments
Zhang et al. [22]	Multimodal Data Processing with GNN	Combines spatial and temporal analysis	Challenging in large-scale networks	Highly relevant for comprehensive IoDT applications
DroneSSL	GNN with Reconstruction and Self-Supervised Learning	Fuses global and local drone temporal features	May require extensive training data	Highly effective for large-scale, dynamic scenarios like bushfire monitoring

The following is a summary of this paper's main contributions.

1) *TinyML-Driven Multimodal Data Analysis:* Our framework leverages the strengths of TinyML and spatial crowdsourcing to process complex multimodal data flows in IoDT systems. By employing lightweight machine learning models and drones for data collection, we ensure efficient processing of data from diverse spatial locations, enhancing spatiotemporal correlation analysis. This approach surpasses traditional methods by integrating the temporal attributes of multimodal multi-drone data and the spatial connections among drones and UGVs, providing a comprehensive overview of environmental conditions.
2) *Lightweight Anomaly Detection with Temporal Feature Extraction and GNN:* Introducing DroneSSL, a novel anomaly detection system that embodies the fusion of spatial crowdsourcing with TinyML. DroneSSL integrates a temporal feature extraction module and a Graph Neural Network (GNN) for advanced spatiotemporal analysis. Optimized for IoDT devices with limited computational resources, this system leverages self-supervised learning and autoencoder methodologies, allowing for the efficient processing of uplinked data from drones and additional sensory data from PoIs through UGVs.
3) *Adaptable and Scalable GNN Architecture:* Our framework addresses the scalability challenge in IoDT by introducing a flexible GNN architecture that can adapt to various GNN kernels, suitable for different anomaly detection scenarios. By clustering drones and employing spatial crowdsourcing for data collection and processing, we propose a strategy for conducting data flow training and inference on selected anchor drones and UGVs. This approach ensures minimal hardware strain on IoDT devices and maximizes the efficiency of TinyML models, crucial for real-time operations in the spatially diverse and resource-constrained environments of IoDT networks.

The structure of this manuscript is as follows: In Section II, the fundamental concepts and relevant studies in anomaly detection are presented. We describe our proposed self-supervised learning anomaly detection approach for the Internet of Drone Things in further depth in Section III. In Section IV, we do an extensive experimental evaluation of our techniques. The study concludes with a summary and concluding remarks in Section V.

II. RELATED WORK

In recent years, the Internet of Drone Things (IoDT) has posed significant challenges in anomaly detection, prompting researchers to propose various solutions. For instance, [14] utilized Euclidean distances to cluster IoDT data points for anomaly detection, but this method did not leverage the spatial correlations between drones. An alternative method proposed by [15] focuses on anomaly detection in multinode data flows using correlation analysis, adeptly managing high-dimensional time series data in IoDT and optimizing computational resources. Yet, this approach does not account for the multimodal nature of IoDT, where each drone is capable of recording data across various modalities.

The escalating scale and intricacy of IoDT, characterized by varied data sources and dynamic threat landscapes, highlight the shortcomings of current anomaly detection techniques. These include limited generalization, and struggles with high-dimensional, heterogeneous, and imbalanced datasets. Deep learning technologies [23] are being progressively employed to surmount these challenges. Research such as [16] has delved into unsupervised multivariate anomaly detection utilizing generative adversarial networks (GAN) coupled with LSTM to discern temporal patterns in complex data streams. Likewise, [17] developed an anomaly detection methodology combining a variational autoencoder (VAE) with LSTM, and [18] introduced a CNN-based framework augmented with incremental learning.

Using spatial properties of network topologies is essential for robust anomaly detection in the Internet of Things. In IoDT networks, adjacency and attribute matrices are rich sources of important characteristics that are well-suited for graph neural networks (GNNs). Important GNN types are GraphSage [24], graph convolution networks (GCNs) [25], and graph attention networks (GATs) [26]. They are all widely used for graph representation learning in IoDT, with the goal of extracting low-dimensional representations from intricate graph structures.

Graph embedding technologies and unsupervised learning play a critical role in IoDT anomaly detection. Unsupervised techniques, reliant solely on normal samples for model training, are bifurcated into reconstruction and predictive modalities. Methods such as [27] employed GCNs for node embedding and data reconstruction, while [28] advanced to GATs. Reference [29] merged structure learning with GNNs for anticipatory anomaly detection, and [30] fused forecast-centric models with reconstruction-centric frameworks to forge an innovative anomaly detection paradigm.

Addressing the challenge of overfitting in traditional unsupervised reconstruction methods, self-supervised learning has emerged as a solution to enhance neural network generalization. In their work, [31] crafted a model predicated on a two-phase adversarial training approach. In parallel, [32] combined GraphSage approaches with the Deep Graph Infomax (DGI) methodology to improve the performance of their model. Other innovative frameworks like [33] explored multiscale adversarial training.

However, these methods typically overlook the IoDT's unique three-dimensional data flow nature, encompassing nodes, modes, and time. To address this, [22] processed multimodal data flows separately, extracting spatial features and using prediction errors for anomaly detection. Despite its effectiveness, the method faced challenges in large-scale IoDT scenarios due to branch expansion with increasing drone numbers.

Combining the benefits of self-supervised learning and reconstruction-based models, our method combines the temporal properties of local and global drones. This integration not only augments detection efficacy but also minimizes the requirements for time and computational resources. This approach is particularly suited for large-scale IoDT applications, such as Australian bushfire monitoring.

III. System Model

A. Problem Formulation

In the domain of the Internet of Drone Things (IoDT), systems typically monitor extensive areas through a network of densely deployed drones. The data collected by a drone and its neighboring drones often show significant correlations due to their spatial proximity. For instance, when a drone detects a bushfire, it is expected that nearby drones will also register changes in their environmental readings, such as a noticeable increase in temperature.

The data collected by drones are not isolated; they are multimodal, indicating that changes in one type of data (e.g., temperature increase) are often accompanied by changes in other types (e.g., increase in CO2 levels and decrease in humidity). This suggests a direct relationship between temperature and CO2 concentration and an inverse relationship with humidity.

The primary aim of this research is to exploit these spatiotemporal correlations for effective anomaly detection in IoDT environments. We approach the anomaly detection challenge as a regression problem focused on forecasting time series data. The regression model for a univariate time series $\{X_t\}_{t\in T}$ is defined as follows:

$$X_{t+W} = f(X_t, X_{t-1}, \dots, X_{t+W-1}|\theta) \tag{1}$$

Here, f represents a mapping function with parameters θ that need to be estimated within the model f. By utilizing the time series $\{X_t\}_{t\in T}$ and considering W as the window length of the timestamp, we aim to create a graph neural network mapping model f. Assuming f as a neural network mapping function (with a single hidden layer for simplicity), our original equation transforms to:

$$X_{t+W} = \theta_0 + \sum_{j=1}^{D} \theta_j g\left(\theta_{0j} + \sum_{i=1}^{W} \theta_{ij} X_{t+i-1}\right) \tag{2}$$

In this equation, D represents the number of nodes in the hidden layer, g is an activation function, and $\theta_0, \theta_j, \theta_{0j}$, and θ_{ij} are all trainable parameters. This graph neural network model, which extends conventional neural network models, uses traditional time-series data for training. The historical data is aligned with current data to identify θ within f, capturing essential characteristics of standard time-series data flow.

For anomaly detection testing and evaluation, the function f predicts data for the current timestamp. An IoDT system is considered to be in an anomalous state at a given time if the prediction error exceeds a predefined threshold.

We define A, a matrix representing spatial relationships among IoDT nodes, and $X_t \in R^{M\times N}$, which consolidates readings from multiple modalities across M drone nodes. Given a timestamp window W, X_t, covering W timestamps, along with A, yields:

$$\tilde{X}_{t+W} = f(X_t, X_{t+1}, \dots, X_{t+W-1}; A|\theta) \tag{3}$$

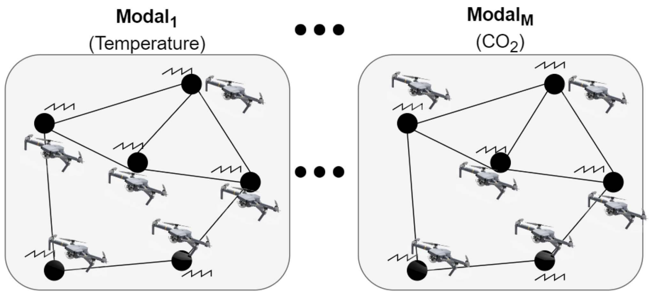

Fig. 1. Illustration of IoDT's multi-drone, multimodal data collection. It shows drones acquiring various types of data, represented by M, such as temperature and humidity.

The parameters (θ) are fine-tuned by minimizing the root mean square error between the predicted matrix $\tilde{X}_{t+W}$ and the actual observations in matrix X_{t+W}. A 'Score' function calculates inference scores, indicating the likelihood of anomalies in the system. A high prediction discrepancy leads to a higher inference score, suggesting potential anomalies. If the inference score at a specific timestamp exceeds a set threshold η, it flags an anomaly. The threshold η is typically determined by applying f to a validation dataset.

This research introduces a graph neural network model f and a novel method for determining its parameters θ, considering the topology of the IoDT system. By integrating data from various nodes, modalities, and time instances, our approach aims to accurately identify anomalies within the IoDT system's data flow.

B. Data Flow Model

In the Internet of Drone Things (IoDT), understanding the interplay between different drones and data types demands a unique IoDT data flow structure. Consider a network in the IoDT spanning a specific area, with N drones equipped with M sensors each, for monitoring environmental variables like temperature, humidity, and CO2 levels. These drones use a time synchronization approach for effective data sharing and collection.

For real-time analysis of both past and current data, the system utilizes a sliding window approach [34]. With the present time marked as t and the window size as W, data from the drones is collected over the interval $\{t-W-1, t-W, \ldots, t-1\}$, forming a tensor $X \in R^{M\times N\times W}$. As new data comes in, the window advances, turning the IoDT's data flow into a dynamic temporal network. In this approach, as seen in Figure 1, every drone gathers time series data on a range of parameters, including oxygen (O_2), CO_2 and temperature (temp) levels.

C. Anomaly Detection Model

Figure 2 presents our IoDT anomaly detection approach, centered on a Graph Neural Network (GNN) autoencoder, building on the concepts of [22], [35]. The encoder f_e and the decoder f_d in this autoencoder have the parameters θ_e and θ_d, respectively. Using dimension d, the encoder converts IoDT data into a hidden layer representation Z. Subsequently, the decoder reconstructs this into X', mirroring the original data's format. The combination of Z and X' then passes through a fully connected network and a softmax function to calculate the probability of anomalies. The key equations are:

$$Z = f_e(X; \theta_e), X' = f_d(Z; \theta_d) \tag{4}$$

$$y = \text{Softmax}\big(FC\big(Z, X'\big)\big) \tag{5}$$

We train our methods using a two-stage process. In order to recreate typical IoDT data patterns, the model first learns the data distribution throughout the whole time series. In the second stage, it is further trained with data incorporating simulated anomalies, enhancing its capacity to differentiate between normal and abnormal patterns at different times. Further discussion on these training strategies appears in the later sections.

D. Preprocessing Module

In the Internet of Drone Things (IoDT), where measurement attributes vary widely, data standardization is essential to maintain consistent scales. Uneven measurement scales can skew analysis, either overstating or understating the importance of certain attributes. To counter this, we adopt Z score normalization for our input tensor, as defined by:

$$\tilde{X}_{ij} = \frac{\big(X_{ij} - \mu\big(X_{ij}\big)\big)}{\sigma\big(X_{ij}\big)} \tag{6}$$

Here, the input tensor is denoted by $X \in R^{M\times N\times W}$ and the data gathered by sensor i on drone j is indicated by X_{ij}. The mean and standard deviation of X_{ij} are $\mu(X_{ij})$ and $\sigma(X_{ij})$, respectively. The multimodal data across drones is adjusted using this Z score normalization approach to a standard distribution with a mean of 0 and a standard deviation of 1, harmonizing the diverse data readings.

To address the dynamic nature of IoDT environments, where drone nodes exhibit mobility and network topology may evolve, our graph neural network (GNN) model is designed with adaptability and scalability in mind. GNNs are particularly well-suited for dynamic networks due to their ability to learn and update node representations based on the changing structure of the graph. In the context of IoDT, as drones move and the network topology changes, the GNN model dynamically adjusts the node embeddings to reflect new spatial relationships and connectivity patterns among drones. This allows the anomaly detection process to remain accurate and reliable even as drones enter or leave the network, move within the environment, or when the data patterns they generate evolve over time.

Furthermore, the model's training process can be periodically updated with new data to refine the graph representations and adapt to long-term changes in the IoDT environment. This continual learning approach ensures that the model remains effective in detecting anomalies in a changing environment. Future enhancements will focus on improving the model's real-time adaptability to sudden changes in network topology and exploring strategies for incremental learning that minimize computational overhead while maintaining detection accuracy.

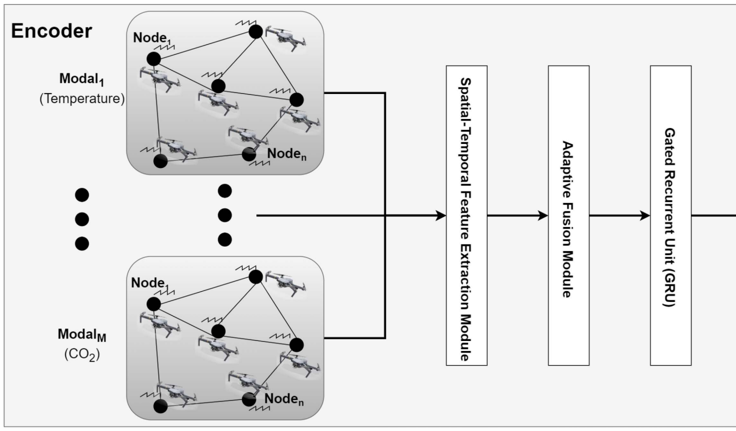

Fig. 2. Overview of the DroneSSL framework, processing IoDT's time series data. It involves encoding this data into a hidden layer Z, then decoding to reconstruct it as X', and finally, employing a fully connected network using both the reconstructed output and encoder's hidden layer to assess anomaly probabilities in the system.

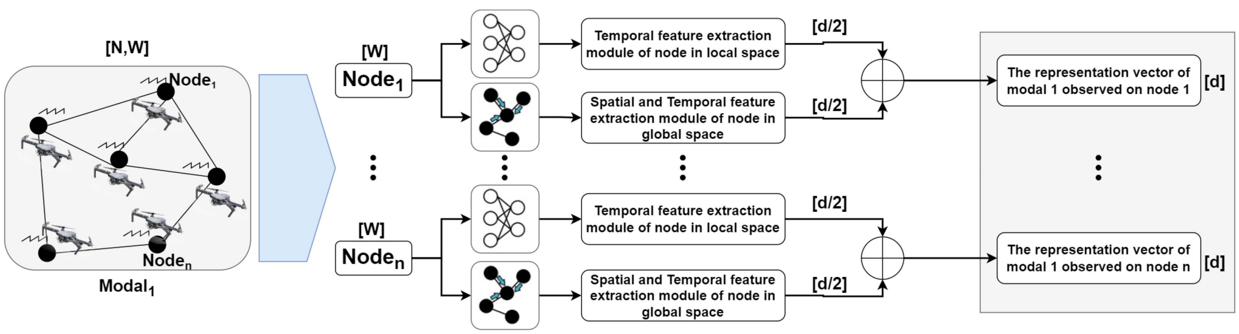

Fig. 3. A module for extracting spatial-temporal features from IoDT data, focusing on a single data type. It extracts temporal and spatial characteristics from multiple drones' data, resulting in a feature matrix that encapsulates the characteristics of multiple drones focusing on a single mode.

E. Encoder and Decoder

The encoder in our IoDT system comprises two primary elements: a module for extracting spatiotemporal features from drones based on individual data types, and an adaptive fusion module for integrating these features.

In the IoDT setup, each data type forms a separate layer, with $X_i \in R^{N\times W}$ representing the data flow for modality i, embodying multinode data streams of that modality. The first step involves processing data through a module dedicated to deriving spatial and temporal features for each drone, specific to a single modality. This results in specialized spatiotemporal feature representations for each drone, respective to the modality. After processing all modal branches, these features are combined using adaptive fusion, facilitating the extraction of intermodal correlations. Then, to handle long-term dependencies, a Gated Recurrent Unit (GRU) is utilized, which serves as the foundation for the encoder's hidden layer. The decoder, structured similarly to the encoder, focuses on reconstructing the original input from this hidden layer. Detailed explanations of these components are provided in the following sections.

1) Spatial-Temporal Feature Extraction Module: As illustrated in Figure 3, this module within the IoDT framework is bifurcated into two parts: one for local temporal feature extraction, and another for global spatial-temporal feature extraction.

Adhering to [19]'s approach for time series feature extraction, our module for local temporal features employs a fully connected neural network. It processes IoDT data X_{ij} (the data from drone j in mode i), a W-dimensional vector, to distill it into a $d/2$-dimensional vector. This is achieved through a series of layers, with K denoting the total number of layers. The operations at each layer k involve the hidden layer output H^k, the initial data vector H^0, the layer weights a^k, biases b^k, and the output function g. The process transforms $X_{ij} \in R^W$ into a reduced feature vector $X^l_{ij} \in R^{d/2}$, as detailed in Equations (7)-(9).

$$H^0 = X_{ij} \tag{7}$$
$$H^k = \sigma\left(a^k H^{k-1} + b^k\right), \quad k \in 1, 2, \ldots, K \tag{8}$$
$$X^l_{ij} = g\left(H^K\right) \tag{9}$$

For global spatial-temporal features, a Graph Neural Network (GNN) is employed. It integrates mode i data X_{ij} from drone j with the spatial topology of the IoDT, considering adjacent drone data to extract common global features. This process reduces the data dimensionality to $d/2$ and is described in Equation (10). Here, X^g_{ij} results from combining adjacent drone data $\{X_{iu}, \{H_{ju}\}\}$ and processing through GNN functions Φ_{GNN} and f_{GNN}. Our framework is adaptable to various GNN types, optimizing anomaly detection in IoDT data.

$$X^g_{ij} = \Phi_{GNN}\left(X_{ij}, f_{GNN}\left(\left\{X_{iu}, \left\{H_{ju}\right\}\right\}\right)\right), \quad u \in N(j) \tag{10}$$

2) Adaptive Fusion Module: The IoDT framework incorporates an adaptive fusion module to integrate unique and common features from different modalities of drones. The procedure initiates by concatenating these distinct and shared attributes, followed by the application of a modality-specific fusion weight q to create a comprehensive feature set for each drone, denoted as $X^c_{:j}$. Equation (11) describes this process, where the fusion weight for the i^{th} modality is $q_i \in R^d$. During training, these originally random weights are adjusted.

$$X^c_{:j} = \sum_{i=0}^{M} q_i \left(X^g_{ij} \| X^l_{ij} \right) \tag{11}$$

To better capture temporal features in IoDT data and address long-term dependencies, the study incorporates dual Gated Recurrent Unit (GRU) layers [36]. This leads to the creation of the encoder's hidden layer representation, denoted by h. As discussed in Section IV-B, an autoencoder network is used for data reconstruction and classification. The decoder, mirroring the encoder's architecture, reconstructs the initial input X' from the hidden layer, as outlined in Equations (12) and (13).

$$Z = GRU\left(X^c, h\right) \tag{12}$$

$$X' = decoder(Z) \tag{13}$$

F. Training

IoDT anomaly detection using deep learning on unlabeled data mainly falls into two categories: unsupervised and self-supervised learning techniques.

Unsupervised Methods: These methods focus on reconstructing normal input data or predicting future values, using the resulting errors for anomaly detection. A key tool here is the autoencoder, designed to encode and then decode input data. Its goal is to learn the data distribution, minimizing reconstruction errors to create a compact representation in its hidden layer. Significant reconstruction errors indicate data anomalies.

Self-Supervised Methods: To avoid overfitting, especially with limited normal data samples, self-supervised learning is used. This technique employs adversarial learning, distinguishing between real (positive) and fake (negative) samples, with negative samples generated through methods like Generative Adversarial Networks (GANs). Self-supervised learning essentially converts an unsupervised task into a supervised one.

Our IoDT framework employs a dual-phase training strategy. It combines the reconstruction efficiency of unsupervised methods with the graph representation benefits of self-supervised learning. The first phase acquaints the model with normal data distribution, while the second involves introducing anomalies to generate negative samples. This is crucial for the model to learn to distinguish between artificial anomalies and regular data.

1) Stage 1: Reconstruction in DroneSSL Training: The first phase of DroneSSL training is focused on reconstructing standard IoDT input data, as shown in (14). In this step, the encoder is used to transform IoDT data X into a hidden layer representation Z, and the decoder is used to reconstruct it. The primary goal is to minimize the reconstruction error. The calculation for the loss function is as follows: l is the total number of elements in X, and x_i and x'_i are the elements in X and X' respectively.

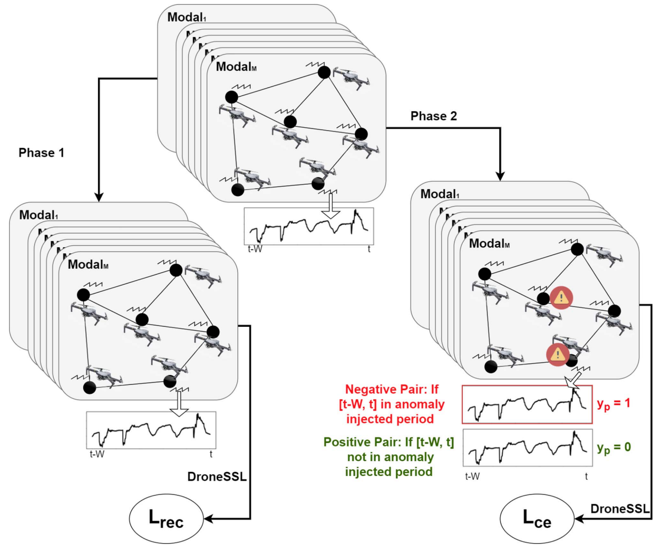

Fig. 4. DroneSSL's two-stage training process: initial reconstruction of standard input, followed by distinguishing anomalies. The visual representation is simplified for clarity, showing data flow for a single modality from one drone.

$$L_{rec} = \frac{1}{l} \sum_{i=0}^{l} \left(x_i - x'_i\right)^2 \tag{14}$$

2) Stage 2: Self-Supervised Learning in DroneSSL: The second phase of DroneSSL training aims to teach the model to identify anomalies in new data. This involves creating negative samples by inserting anomalies into normal data. The primary goal is to distinguish between abnormal and normal data. Equation (15) defines the loss function, which aims to reduce classification error. It takes two inputs: the projected probability of having a positive sample (y_p) and the actual label (y.

$$L_{\text{ce}} = -y \log\left(y_p\right) - (1 - y) \log\left(1 - y_p\right) \tag{15}$$

Figure 4 illustrates DroneSSL's two-stage training process. The training combines elements from (14) and (15) into an overall training formula, given in (16). The number of training epochs is indicated by n in this case. In order to evaluate the present state of the system, the binary classification probability is calculated during the inference phase using the classification output branch.

$$L = \frac{1}{n} L_{rec} + \left(1 - \frac{1}{n}\right) L_{\text{ce}} \tag{16}$$

G. Enhancing Scalability: Strategy for Large-Scale IoDT Scenarios

In IoDT applications with large datasets, the computational and storage requirements increase significantly, especially with the complex layers of Graph Neural Networks (GNNs). This research presents an enhanced DroneSSL technique, termed DroneSSL+, that integrates Piecewise Aggregate

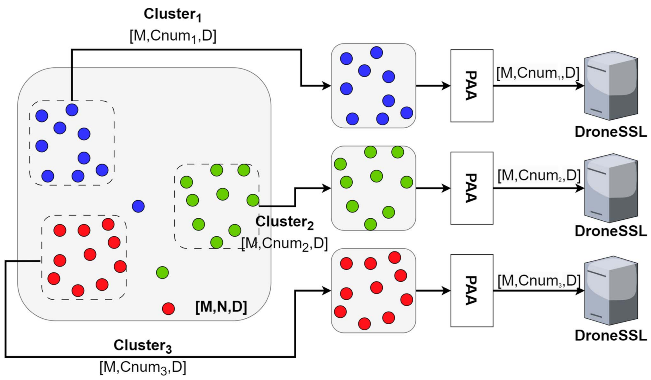

Fig. 5. DroneSSL$^+$ optimization strategy in a large-scale IoDT scenario.

Approximation (PAA) with K-means clustering to overcome these issues in large-scale IoDT.

K-means clustering, known for its efficiency and simplicity, segments drones into clusters based on their spatial coordinates C and a user-defined cluster number k, using Euclidean distances. PAA, as per [37], is effective for reducing data dimensionality while retaining key attributes. It compresses a temporal data sequence $Y = \{Y_1, Y_2, \ldots, Y_n\}$ of length n into a shorter sequence $\bar{Y} = \{\bar{Y}_1, \bar{Y}_2, \ldots, \bar{Y}_m\}$ of length m, as shown in (17).

$$\overline{Y_i} = \frac{m}{n} \sum_{j=\frac{n}{m}(i-1)+1}^{\frac{n}{m}i} Y_j, \quad m \leq n \tag{17}$$

Figure 5 displays the initial step of DroneSSL+, where drones are grouped into k clusters via K-means based on their locations. Each drone is positioned closer to its cluster center than to others. Afterwards, the temporal data dimension is reduced using PAA. The segmented dataset then undergoes independent training and inference on separate devices. This division in DroneSSL+ significantly reduces model parameters, enabling parallel processing on multiple devices and improving efficiency. The temporal dimension reduction converts D data points into fewer points (D_1), thus lessening the input data points and preserving essential information, thereby enhancing system speed.

H. Ethical Considerations and Privacy

In the deployment of DroneSSL for anomaly detection within the Internet of Drone Things (IoDT), ethical considerations and privacy concerns take on paramount importance, especially given the system's potential application in sensitive and diverse environments. The utilization of drones equipped with various sensors for monitoring purposes raises significant privacy issues, as these devices can inadvertently collect personal data without consent. To mitigate such risks, our framework incorporates strict data handling and processing protocols that align with established privacy regulations, such as the General Data Protection Regulation (GDPR) in the European Union. These protocols ensure that all data collected by drones are anonymized and encrypted, preventing the identification of individuals from the sensor data. Additionally, DroneSSL employs mechanisms to limit data collection to strictly necessary parameters, avoiding the capture of irrelevant personal information. Furthermore, the deployment of DroneSSL involves stakeholder engagement and transparency measures, ensuring that individuals in monitored areas are informed about the data collection practices and their rights. By prioritizing ethical considerations and privacy protection, DroneSSL aims to foster trust and acceptance among the public, ensuring its ethical deployment in diverse environments while safeguarding individuals' privacy rights.

IV. Analysis and Discussion

To validate the effectiveness of our anomaly detection approach for the Internet of Drone Things (IoDT), several experiments were conducted. For the experimental setup, we utilized sensor data from the Internet of Drone Things (IoDT). The hardware configuration included an AMD Ryzen 7 3700X CPU running at 3.6 GHz, coupled with an NVIDIA GTX 1080 Ti GPU. The software framework was based on Python, with PyTorch-1.9.0 serving as the primary tool for model implementation and testing of various approaches. Data visualization was managed using Python's matplotlib library. For enhanced computational efficiency, GPU acceleration was enabled through CUDA version 11.2.

A. Dataset

Our model's validation employed the dataset from the Intel Berkeley Research Lab (IBRL) [38], adapted to an IoDT context. The original IBRL network had 54 sensors measuring temperature, humidity, illumination, and voltage from 28 February to 5 May 2004.

In our experiments, we used data from 51 drones, capturing three modalities (temperature, humidity, and voltage) over 5000 moments between March 4 and March 9. This created a $50 \times 3 \times 5000$ data tensor. The training and test sets were split 6:4, with 3000 time points for training and 2000 for testing, covering three modalities across 51 drones. DroneSSL's second stage employed self-supervised learning, including negative sample creation through anomaly injection.

B. Anomaly Injection Methods

We tested five anomaly detection methods:

1) Scale Change: Multiplying IoDT data values by random constants within the sliding window.
2) Negation: Inverting IoDT data values by a factor of -1 to create mirrored data.
3) Sudden Change: Significantly altering IoDT data values, as shown in (18).

$$X_{ij}(t) = X_{ij}(t) \pm \left(X_i^{\text{up}} - X_i^{\text{down}}\right), t \in \{1, 2, \ldots, \tau\} \tag{18}$$

4) Intermodal Anomaly: Introducing anomalies that disrupt correlations between different modalities.
5) Internode Anomaly: Injecting anomalies to break correlations in a single mode among multiple drones.

These experiments aimed to evaluate DroneSSL's anomaly detection capabilities in a complex IoDT environment, adapting traditional wireless sensor network methods to IoDT.

C. Performance Assessment Metrics for DroneSSL

For evaluating the efficacy of the DroneSSL method in detecting anomalies within the IoDT framework, we focus on three primary metrics: the F1 score (F1), recall (Rec) and precision (Prec), all calculated on the test dataset.

Prec (Precision): Precision is the ratio of correctly predicted abnormal samples to all samples that DroneSSL expected to be abnormal.

Rec (Recall): Recall assesses the percentage of correctly identified abnormal samples by DroneSSL compared to the actual abnormal samples present in the dataset.

F1 Score: Precision and recall are seen in balance by the F1 score, which is a harmonic mean of the two. It's derived from the confusion matrix elements: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). TP indicates accurately detected anomalies, FP represents incorrect anomaly detections, TN counts correct identifications of normal instances, and FN refers to anomalies that were not detected. The formula for F1 is $F1 = 2 \times \frac{\text{Prec}\times\text{Rec}}{\text{Prec}+\text{Rec}}$.

For testing, the test set is split into two subsets: one with timestamps for anomaly injection (set-1) and another without anomalies (set-2). Every test set undergoes a total of T anomaly detection iterations. In each iteration, one of the four anomaly injection methods is applied at a random timestamp in set-1. Anomaly detection by DroneSSL is considered accurate (TP) if it occurs within $[t, t + delaystep]$ after an anomaly is injected, where t is the timestamp and *delaystep* is the allowed detection delay, set equal to the sliding window length W. Missed detections are marked as FN.

For set-2, DroneSSL correctly identifying a normal point as normal is labeled *TN*, and wrongly predicting an anomaly is considered *FP*.

D. Sensitivity Analysis of the Model

To assess our model's sensitivity to abnormal data, we initially trained it using standard parameters and then tested its response to anomalies with varying deviations from normal data.

The outcomes of this sensitivity analysis are shown in Figure 6. As previously mentioned in Section III, the deviation factor for internode and intermodal anomalies is represented by the symbol p. An increase in p signifies a smaller deviation from normal data. The graph indicates that as p rises, the model's recall diminishes, suggesting the model is more adept at distinguishing significant anomalies (lower p values) from normal data.

After establishing the model's capability to differentiate normal and abnormal data, we fine-tuned its parameters to optimize performance. The sliding window length W was adjusted within the range {10, 15, 20, 25, 30}, the learning rate lr was set within the range {0.00001, 0.00005, 0.0001, 0.0005}, the GRU hidden layer size was adjusted with the range {8, 16, 32, 64}, and the anomaly deviation control variable p was set to 40. The anomaly injection duration ($t_e - t_s$) was fixed at 10. DroneSSL and baseline methods underwent multiple tests within these parameters to derive the best results, which are detailed in the following sections.

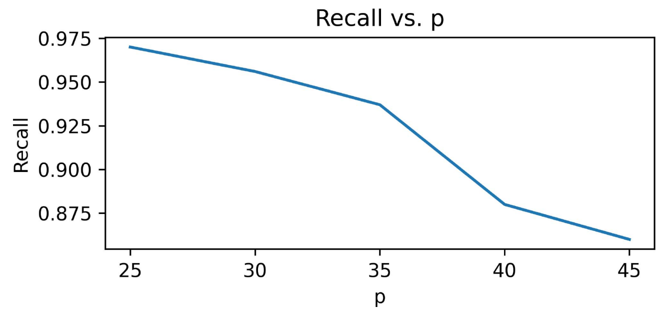

Fig. 6. Graph depicting the sensitivity test results, where a higher p value corresponds to a smaller anomaly deviation from normal data.

E. Baseline Method Comparisons for DroneSSL

We examined our DroneSSL approach against three deep learning-based anomaly detection techniques in order to assess its effectiveness in the IoDT environment.

1) CNN-LSTM Model: An 8-layer CNN-LSTM model was built using PyTorch, comprising a sequence of Conv2d, MaxPool2d, LSTM, and FC layers. This model classifies normal IoDT data as class 0 and data altered by anomalies as class 1. Its four-dimensional tensor input, matching the IoDT data structure, is formatted as [batch, modal_num, drone_num, window_size]. However, unlike GNNs, CNNs lack the capability to exploit the inherent topological features of IoDT networks.

2) Anomaly Detection Using Multivariate Time-Series Data: Zhao et al. [30] introduced a model combining prediction-based and reconstruction-based approaches. This method employs dual Graph Attention Networks (GATs) to extract correlation features between modes and temporal features over time. After concatenation, these features are processed through a GRU, and the network bifurcates for next-timestamp prediction and input data reconstruction. The model, initially designed for single-node multimodal data, was adapted to our multi-drone scenario. For our dataset, 51 separate Zhao et al. [30] models were developed, one for each drone, with performance metrics representing an average of these models' results.

3) Anomaly Detection Using Multinode Multimodal Time-Series Data: Zhang et al. [22] extended Zhao et al. [30]'s approach by including spatial location information of multiple drones. The model uses multiple GAT branches to extract features from various modalities and time sequences across drones. A GRU is utilized for long-term dependencies, and a GAT captures the spatial position of drones. It predicts next-timestamp values for all modes and drones, with anomalies identified when the prediction error exceeds a certain threshold. When anomalous data greatly deviates from the usual data distribution, this approach works especially well.

Table II presents the experimental results comparing the performance of DroneSSL with other baseline methods in IoDT anomaly detection. The results include precision (Prec), recall (Rec), F1 score, and accuracy (Acc) for each method.

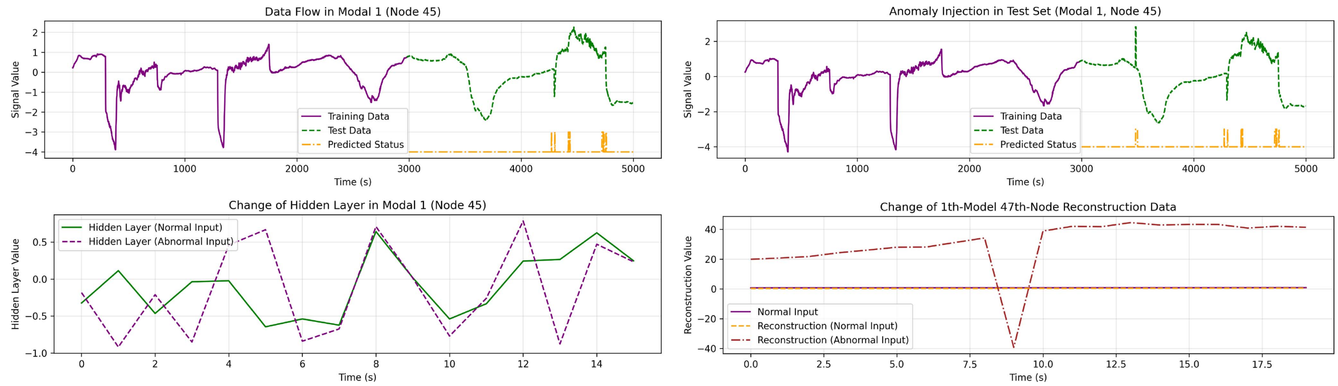

Fig. 7. Point anomaly test visualization for DroneSSL.

TABLE II
COMPARISON OF ANOMALY DETECTION METHODS IN IoDT

Method	Prec	Rec	F1	Acc
CNN-LSTM	78.3%	71.2%	74.6%	75.5%
Zhao et. al. [30]	76.8%	85.3%	80.9%	80.2%
Zhang et. al. [22]	88.4%	81.7%	84.9%	85.6%
DroneSSL (GCN)	**96.2%**	82.5%	88.7%	88.9%
DroneSSL (GAT)	93.7%	**86.3%**	**89.6%**	**90.4%**

The CNN-LSTM method exhibited lower performance metrics, primarily due to its limited ability to exploit the IoDT network topology compared to GNNs. GNNs enable drones to derive information from all adjacent drones, which proves more effective than the CNN's convolutional layers confined to the kernel's range. Zhao et al. [30], focusing on multimodal data correlations, does not incorporate spatial location features of drones, requiring separate model training for each drone. Furthermore, Zhao et al. [30]'s model sometimes experiences training difficulties like gradient explosions, affecting its performance relative to Zhang et al. [22] and DroneSSL.

Zhang et al. [22]'s method, though utilizing GNNs and GRUs for timestamp-based feature extraction, struggles with long-term dependencies, especially in scenarios with larger training sets than test sets. This limitation can impact its ability to detect distant correlations, reducing its anomaly detection accuracy.

In contrast, DroneSSL merges reconstruction-based models with self-supervised learning for effective normal data feature capture and graph representation learning generalization. It employs GNNs for both global common node information extraction and individual drone data analysis. DroneSSL supports various GNN types, with both GCN and GAT-based approaches outperforming others. Specifically, GAT-based DroneSSL achieves the highest overall performance, with its F1 score surpassing Zhao et al. [30] by 4.9% and Zhang et al. [22] by 8.7%.

F. Interpretability of DroneSSL

We ran a visual analysis on training and test data to confirm that DroneSSL can comprehend normal data distribution and detect anomalies. This involved looking at the reconstructed data X' and the hidden layer representation Z for a particular drone.

Point Anomaly Test: For the second modality from drone 47, an analysis of raw data over 5000 time points was performed, as Fig. 7(a) illustrates. Test data is represented by the green curve, while training data is shown by the purple curve. Predicted values are displayed on the orange curve, with a possible anomaly indicated by a spike. Notably, the prediction curve's multiple spikes to 1 might suggest either misjudgments or unrecognized anomalies in the original data. The response of DroneSSL to a purposefully created point anomaly at the 500th moment in the test set is seen in Fig. 7(b), which confirms the model's capacity to identify significant deviations from normal ranges by displaying a marked rise in judgment values. Fig. 7(c) displays the alterations in the hidden layer vector for the same modality and drone around the 511th moment, where an anomaly was introduced. The green line indicates the vector without anomalies, while the purple line shows post-anomaly alterations, revealing a distinct difference. Fig. 7(d) presents the reconstruction curves with and without an anomaly at the 510th moment. The normal input in this case is shown by the purple curve, its reconstruction is shown by yellow, and the reconstruction post-anomaly is shown by brown. A notable numerical shift in the reconstruction curve after the anomaly injection demonstrates DroneSSL's sensitivity to abrupt deviations from normal data. These visual assessments affirm DroneSSL's efficiency in learning normal data patterns and its adeptness in spotting and responding to anomalies within the IoDT framework.

Contextual Anomaly Test: As seen in Fig. 8(a), we examined 5000 minutes of data from drone 45's first modality in order to further illustrate DroneSSL's capacity to identify contextual abnormalities. The orange curve highlights anticipated anomalies, whereas the purple curve shows training data and the green curve indicates test data. DroneSSL's response to contextual anomalies inserted between test set moments 700–1000, while adjacent drones exhibit similar declines, is shown in Fig. 8(b). A subtle deviation from normal ranges was introduced at the 850th moment, which DroneSSL successfully identified as a contextual anomaly. Fig. 8(c) depicts changes in the hidden layer vector for drone 45 around the 860th moment, with the green line showing the vector without anomaly and the purple line post-anomaly injection, highlighting a significant difference. Fig. 8(d) shows the reconstruction curves

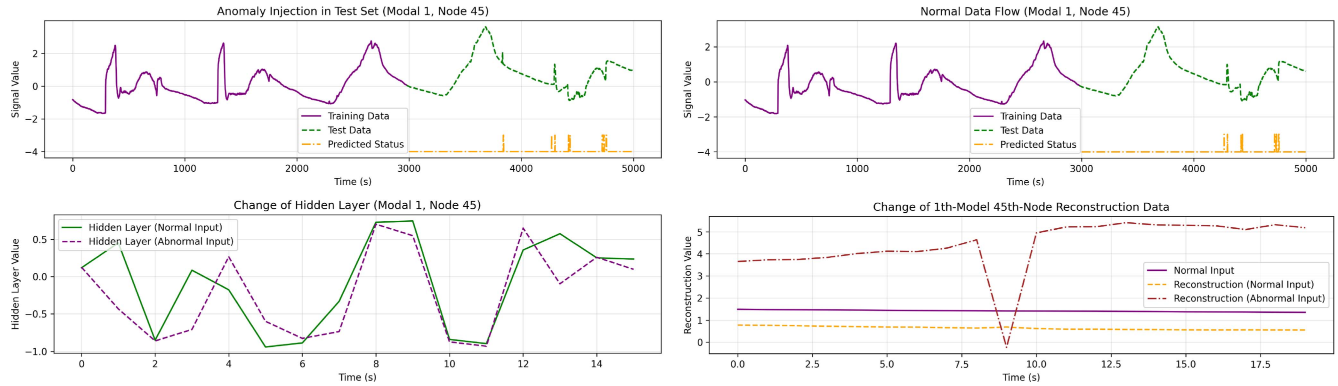

Fig. 8. Visualization of contextual anomaly detection in DroneSSL, showing data and prediction curves for drone 45.

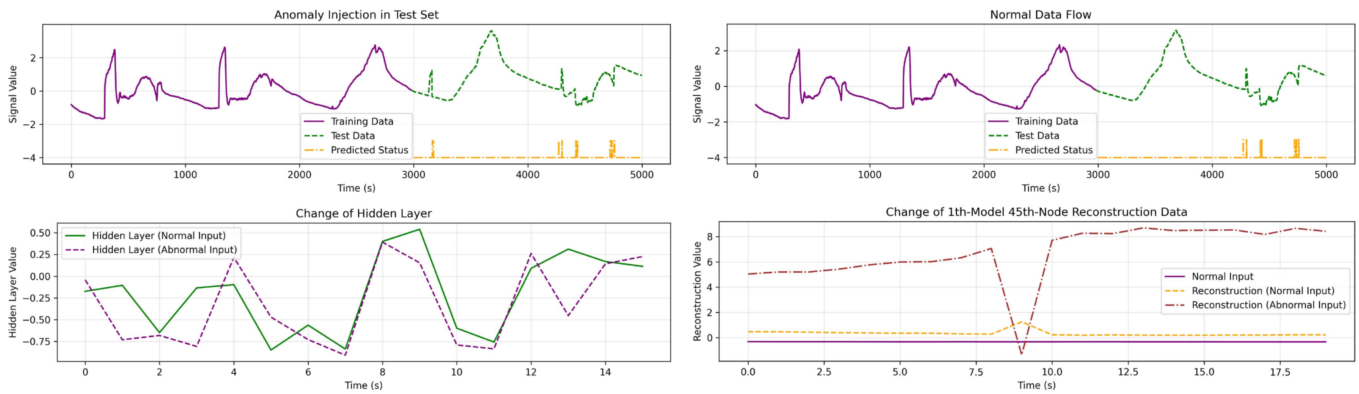

Fig. 9. Periodic anomaly test visualization for timestamps 150-180, showing DroneSSL's response to injected anomalies.

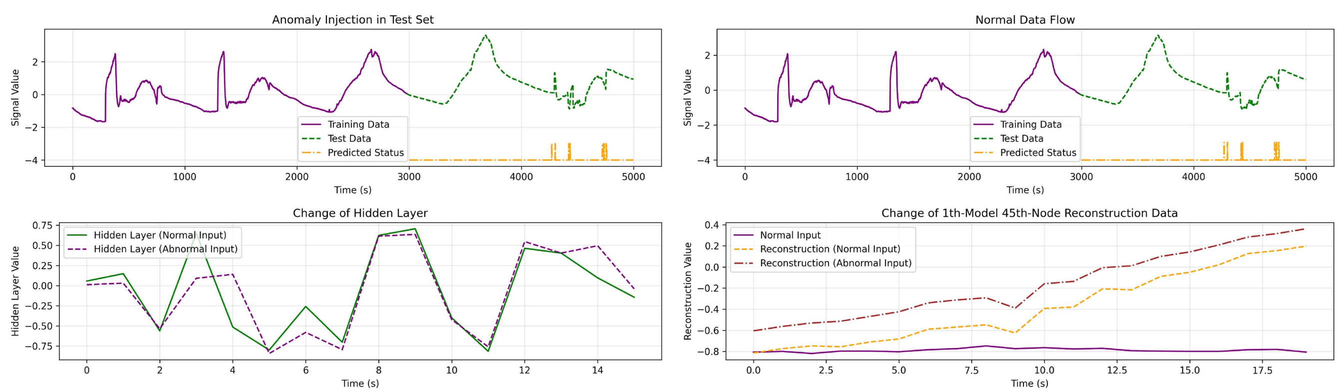

Fig. 10. Periodic anomaly test visualization for timestamps 1450-1480, highlighting DroneSSL's ability to recognize periodic patterns in IoDT data.

before and after injecting the contextual anomaly at the 860th moment. The numerical change in the reconstruction curve, though less drastic than in point anomaly cases, confirms DroneSSL's sensitivity to subtle deviations from normal patterns.

Periodic Anomaly Test: This test evaluated DroneSSL's understanding of periodic normal data patterns. In the test set data of drone 45, a unique form that resembled those in the training set was added between moments 150-180 and 1450-1480. The results of the specific shape injection into moments 150–180 are shown in Fig. 9. This shape introduces a large anomaly, as shown in Fig. 9(b), which is accurately detected by DroneSSL during the lag period. The hidden layer vector and reconstruction curve changes at timestamp 183 are displayed in Fig. 9(c) and (d), respectively, illustrating DroneSSL's capability to identify and respond to the anomaly. Fig. 10 displays outcomes of injecting the shape into moments 1450-1480. As seen in Fig. 10(b), since the injected shape resembles the training set waveform and aligns with its periodicity, DroneSSL does not classify it as an anomaly. This response, along with the earlier results, showcases DroneSSL's proficiency in discerning periodic patterns in IoDT data. Fig. 10(c) and (d) depict the hidden layer vector and reconstruction curve changes at timestamp 1483, illustrating slight deviations but within acceptable ranges, leading DroneSSL to not flag this instance as anomalous. These tests highlight DroneSSL's capacity to detect anomalies while respecting the inherent periodic nature of normal IoDT data.

TABLE III
LARGE-SCALE IMPROVEMENT STRATEGY WITH DRONESSL^{+}

Experiment	Cluster Number	PAA	F1 Score	Average Running Time (s)
EXP. 1	1	None	89.4%	18.7
	2	None	88.6%	17.2
	3	None	86.3%	15.8
	4	None	84.2%	14.1
EXP. 2	3	None	86.3%	15.8
	3	4/5	87.8%	15.0
	3	3/5	79.1%	11.3
	3	2/5	70.4%	7.5

G. DroneSSL^{+} in Large-Scale IoDT Scenarios

In response to the computational and memory demands of expanding Internet of Drone Things (IoDT) networks, we introduce an advanced version of DroneSSL, termed DroneSSL^{+}. This version integrates K-means clustering and Piecewise Aggregate Approximation (PAA) for enhanced efficiency, as discussed in Section IV-F.

Two sets of experiments were conducted to assess DroneSSL^{+}'s time complexity performance. The first set (Experiment 1) involved clustering the IoDT network and running the DroneSSL framework on separate devices per cluster, reducing model parameters and GPU usage. The second set (Experiment 2) utilized PAA to compress the time dimension of the dataset while maintaining a constant number of clusters, performing anomaly detection at specified intervals to further cut down on runs and time overhead.

The results, shown in Table III, reveal that increasing the number of clusters leads to a slight decline in DroneSSL^{+}'s performance but significantly reduces the average testing time. Notably, reducing the dataset size to 4/5 through PAA improves performance, likely due to PAA's efficiency in preserving general data trends while eliminating some irregularities. However, more substantial reductions, such as to 2/5 of the original size, result in a marked performance drop, with the F1 score falling to 70.4% despite a substantial decrease in average testing time.

Therefore, striking a balance between performance and testing time efficiency is crucial, depending on the specific IoDT application needs. Selecting an optimal number of clusters and PAA reduction ratio is key to successfully implementing DroneSSL^{+} in large-scale IoDT anomaly detection scenarios.

V. CONCLUSION AND FUTURE WORK

To sum up, this paper has introduced DroneSSL, an innovative anomaly detection framework that seamlessly integrates spatial crowdsourcing with the Internet of Drone Things (IoDT) and TinyML to address the complexities of environmental monitoring, notably in scenarios like Australian bushfire management. By utilizing drones and unmanned ground vehicles for data collection across inaccessible terrains, DroneSSL embodies a cutting-edge approach that leverages lightweight machine learning models for efficient and effective spatiotemporal analysis. This framework's combination of temporal feature extraction and a scalable Graph Neural Network (GNN) architecture optimizes anomaly detection in IoDT, demonstrating a significant improvement in performance metrics. DroneSSL's successful application in challenging environments not only highlights its potential in critical monitoring tasks but also underscores the transformative impact of merging spatial crowdsourcing with advanced computational technologies. Looking ahead, the exploration of further optimizations and the expansion of DroneSSL's application scope promise to enrich the IoDT landscape, paving the way for groundbreaking developments in IoT edge computing and environmental surveillance technologies.

Future enhancements to DroneSSL could significantly enhance its efficacy and extend its applicability across a wider range of Internet of Things (IoT) environments, addressing critical challenges such as computational efficiency and real-time data processing. By integrating edge computing and federated learning, DroneSSL could offer decentralized anomaly detection capabilities, reducing latency and bandwidth usage while ensuring data privacy and security through advanced encryption techniques. Additionally, expanding its adaptability to various IoT domains, including industrial IoT (IIoT) for predictive maintenance and smart cities for urban monitoring, necessitates the development of flexible, generalized models that can effortlessly learn from diverse data types and sources. Incorporating transfer learning could further streamline DroneSSL's deployment across different scenarios, minimizing the dependency on extensive labeled datasets. Overcoming regulatory and standardization hurdles, especially in drone airspace management, alongside ensuring robust privacy and security measures, will be paramount for DroneSSL's broader implementation. Such advancements promise to solidify DroneSSL's position as a versatile, efficient tool for monitoring and surveillance within the ever-evolving IoT landscape, offering promising prospects for future research and application in this domain.

Enhancing the validation of DroneSSL's performance further, future work could explore a broader array of datasets, specifically those capturing a wider range of real-world IoDT applications, such as urban surveillance, environmental monitoring, and industrial inspections. Integrating datasets with varying characteristics and challenges, such as different scales of IoDT deployments and diverse anomaly types, would provide a more robust evaluation of DroneSSL's adaptability and efficacy across different scenarios. Additionally, incorporating a wider range of performance metrics, such as detection latency, scalability, and computational efficiency, alongside precision, recall, and F1 score, would offer a more comprehensive assessment of the system's overall performance. A comparative analysis with industry standards and state-of-the-art methods, particularly those deployed in real-world IoDT applications, would further elucidate DroneSSL's competitive advantages, shedding light on its practical implications and potential for widespread adoption.

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